TOPIC 3 · INTERACTIVE DASHBOARDS WITH CUSTOM SKILLS
LAB 8 THEME · CUSTOM DASHBOARD SKILL

Lab 8 — Build an Interactive Dashboard with a Custom Skill

Maps to	LO3: build and validate an interactive Excel maintenance dashboard
Duration	35 minutes
Tools	Excel Copilot · Microsoft 365 Copilot web · OneDrive (Skills folder only)
Mock data	data/Northstar_Work_Orders.xlsx
Lab files	data/templates/water-service-dashboard/SKILL.md (Reviewed interactive dashboard Skill fallback)
Course	GenAI for Engineers · v2.4
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-8

Goal

Install the water-service-dashboard skill and use it to build an interactive dashboard.

What You Will Build

Northstar_Dashboard.xlsx with six KPI tiles, three charts and slicers.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac. Download this lab folder and open the files straight from its data folder; save any new file in the same folder. All data is fictional.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Upload the skill to OneDrive. In Excel open Copilot and select Settings (...) > Manage plugins & skills > Custom skills. Turn Custom skills on and select Add in OneDrive (or Create OneDrive folder, then Open skills folder). OneDrive opens at your skills folder. Select Create or upload > Folder upload and choose the water-service-dashboard folder in this lab's data/templates folder.

PROMPT / CHECK — copy and adapt

```
OneDrive : Documents > Copilot > Microsoft Excel > skills > water-service-dashboard > SKILL.md
```

Step 2. Turn the skill on. Back in Excel select Settings (...) > Manage plugins & skills > Custom skills > Refresh, then turn water-service-dashboard on. Close the dialog and type @wat in the Copilot prompt box: water-service-dashboard should appear. If not, wait a minute and select Refresh again.

PROMPT / CHECK — copy and adapt

```
Check : water-service-dashboard appears when you type @
```

Step 3. Plan the dashboard. Open data/templates/Northstar_Dashboard_Starter.xlsx and save it as Northstar_Dashboard.xlsx. Select New chat, choose Plan and send the prompt. Read Copilot's plan.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@water-service-dashboard Plan a dashboard for tblWorkOrders with six KPI tiles, three charts
(by status, by job category and by site) and slicers. Don't change anything yet.
```

Step 4. Build it. Choose Allow editing, send the prompt and select Done. The tiles should show 120 total, 60 backlog, 50 overdue, 48 closed, on-time 62.5% and cost S$35,827.50.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@water-service-dashboard Build the dashboard you planned, with formulas that read
tblWorkOrders. Don't change the data.
```

Step 5. Polish the charts. Send the prompt. Select Done to keep the change, or Undo to reverse it. Each chart title now states a finding.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@chart-design Improve the three dashboard charts with finding-style titles and data labels.
Don't change any values.
```

Step 6. Check the numbers. Choose Chat only and send the prompt. Replace any typed-in number it finds with a formula, then save.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
Check every KPI tile and chart on the Dashboard against tblWorkOrders. List any number that is
typed in instead of calculated, and any total that does not match.
```

Copy buttons for every prompt: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-8

Test It

Check	Expected result
Tiles	120; 60; 50; 48; 62.5%; S$35,827.50.
Charts	Status, category and site charts each add up to 120; titles state a finding.
Formulas	No typed-in numbers; the source data is unchanged.

Troubleshooting

If this happens	Fix
Copilot cannot see my file	In Excel, save the file and open Copilot again. In Word, PowerPoint or Outlook, upload the file with + > Upload images and files, and upload it again after you change it.
A number is different from the lab	Clear any slicers or filters, check the table has no extra rows, then ask again in a new chat.
A button or feature is missing	Check you are signed in with your course account and Microsoft 365 is up to date, then tell your trainer.

Challenge

Ask Microsoft 365 Copilot on the web to write your own dashboard SKILL.md and compare it with the supplied one.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?